



DPUFlipFlop

Simulates a single flip-flop element.

1. Inputs:

<i>Toggle</i>	Value of flip-flop is changed during transition from 0 to >0.
<i>Set</i>	When this input is <i>active</i> , the flip-flop is set.
<i>Reset</i>	When this input is <i>active</i> , the flip-flop element is reset.
<i>Mode</i>	Mode=0: The flip-flop is level triggered, that is <i>Set</i> and <i>Reset</i> are effective, while they are >0. (Default) Mode=1: The flip-flop is edge-triggered, that is <i>Set</i> and <i>Reset</i> are only effective on a transition from 0 to >0.

2. Dynamic parameters:

<i>Multiplier</i>	Multiplier for computed output values. (Default: 1)
<i>Offset</i>	Offset for computed output values. (Default: 0)

3. Outputs:

<i>Out</i>	Output value
<i>OutNot</i>	Output value for the opposite value of the flip-flop.

4. Functional principle:

DPUFlipFlop simulated a single binary flip-flop element, whose state (1 or 0) can be set using the inputs *Toggle*, *Set* and *Reset*.

We assume z is the state of the flip-flop. Then the output values are calculated as following:

$$Out = z * Multiplier + Offset$$

$$OutNot = (1-z) * Multiplier + Offset$$